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EduVision AI (A Multi-Agent Multimodal Education System)

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Problem Statement & Objectives

Problem Statement:

Modern education systems are at a crossroads. While information is abundant, personalized and engaging delivery remains a bottleneck. Traditional Learning Management Systems (LMS) are static repositories of data that lack:

1. **Contextual Intelligence:** They cannot research and update their knowledge in real-time.
2. **Multimodal Engagement:** Most platforms rely heavily on text, leading to lower cognitive retention.
3. **Autonomous Coordination:** They require manual intervention to curate content, generate assessments, and manage doubt resolution.
4. **Scalable Memory:** Local storage solutions fail to provide a consistent learning history across different devices and large-scale deployments.

The core problem addressed in this project is:

How can a Multi-Agent AI system collaboratively manage teaching, learning assistance, evaluation, and personalized guidance by autonomously coordinating multiple intelligent agents to improve the overall educational experience?

The Multi-Agent Education System aims to solve this problem by designing specialized agents such as tutoring agents, assessment agents, recommendation agents, and monitoring agents that work together to provide adaptive and intelligent learning support.

Project Objectives:

Agentic Orchestration: To develop a "Coordinator" framework that handles complex educational tasks by delegating them to specialized AI agents.

Multimodal Content Generation: To bridge the gap between text and visuals by integrating AI image generation (Text-to-Image) into the learning flow.

High-Speed Execution: To implement parallel processing techniques that generate multiple educational assets simultaneously, ensuring a sub-30-second response time.

Cloud-Native Knowledge Base: To transition from local file-based storage to a professional Cloud Vector Database (Upstash) for semantic search and global persistence.

User-Centric Experience: To design a premium, distraction-free interface that uses modern aesthetics (Glassmorphism, Nebula glows) to improve focus.

Scope of the Project

The Multi-Agent Education System focuses on designing and implementing an intelligent educational environment where multiple AI agents collaborate to assist learners and improve the learning process. The following functionalities are included within the scope of the project:

- Development of a simulated multi-agent educational platform.
- Implementation of specialized agents such as Teaching Agent, Assessment Agent, Recommendation Agent, and Monitoring Agent.
- Autonomous interaction and coordination between agents for task execution.
- Personalized learning assistance based on user inputs and learning progress.
- Automated doubt resolution and concept explanation.
- Student performance evaluation through quizzes or simulated assessments.

Out of Scope:

- Integration with real-time university or school databases.
- Deployment in large-scale production educational institutions.
- Real-time classroom monitoring using IoT or video analytics.
- Advanced adaptive learning using large real-world datasets.
- Full distributed cloud-based multi-agent deployment.

Project Focus:

The project primarily demonstrates how multiple intelligent agents can collaboratively manage educational tasks and provide adaptive learning support within a controlled and simulated environment. The emphasis is on showcasing core Agentic AI concepts such as autonomy, coordination, reasoning, and modular system design.



Proposed Solution

1. Key Features:

- 2. Real-Time Research (Researcher Agent):** Unlike static bots, EduVision AI uses DuckDuckGo Search API to fetch the most recent and credible data from the web.
- 3. Structured Pedagogical Writing (Writer Agent):** Converts raw research into formatted modules, learning objectives, and practical case studies using Llama 3.3 70B.
- 4. Multimodal Flashcards (Vision Agent):** Automatically identifies key concepts and generates 3D scientific illustrations using Pollinations AI, making abstract ideas tangible.
- 5. Asynchronous Parallel Imaging:** Uses Python's ThreadPoolExecutor to generate 5+ high-fidelity images at once, eliminating the sequential wait-time.
- 6. Semantic Memory (Upstash Vector DB):** Every topic searched is converted into high-dimensional vectors and stored in the cloud. This allows the system to "remember" and instantly retrieve complex data without re-processing.
- 7. Self-Correcting Assessments (Quiz Agent):** Generates logic-based MCQ assessments with instant feedback and detailed explanations for every answer.

8. Overall Architecture / Workflow:

- 9. Real-Time Research (Researcher Agent):** Unlike static bots, EduVision AI uses DuckDuckGo Search API to fetch the most recent and credible data from the web.
- 10. Structured Pedagogical Writing (Writer Agent):** Converts raw research into formatted modules, learning objectives, and practical case studies using Llama 3.3 70B.
- 11. Multimodal Flashcards (Vision Agent):** Automatically identifies key concepts and generates 3D scientific illustrations using Pollinations AI, making abstract ideas tangible.
- 12. Asynchronous Parallel Imaging:** Uses Python's ThreadPoolExecutor to generate 5+ high-fidelity images at once, eliminating the sequential wait-time.
- 13. Semantic Memory (Upstash Vector DB):** Every topic searched is converted into high-dimensional vectors and stored in the cloud. This allows the system to "remember" and instantly retrieve complex data without re-processing.
- 14. Self-Correcting Assessments (Quiz Agent):** Generates logic-based MCQ assessments with instant feedback and detailed explanations for every answer.

Tools & Technologies Used:

The implementation of the Multi-Agent Education System may use the following technologies:

Core Logic: Python 3.11 with LangChain for agentic chains.

LLM Engine: Groq Cloud (Llama 3.3 70B) for 10x faster reasoning compared to standard GPT models.

Vector Engine: Upstash Vector DB (Serverless Cloud) for semantic indexing.

Image Gen: Pollinations AI (Stable Diffusion architecture) for watermark-free educational visuals.

Embeddings: sentence-transformers (all-MiniLM-L6-v2) for 384-dimensional vector mappings.

UI Framework: Streamlit with Custom CSS (Nebula Theme).

These technologies support modular development, agent communication, and experimentation with intelligent educational workflows.

Results & Output:

1. Screenshots / Outputs

The Multi-Agent Education System generates structured and meaningful outputs that demonstrate how multiple agents collaboratively assist the learning process. The system produces results such as:

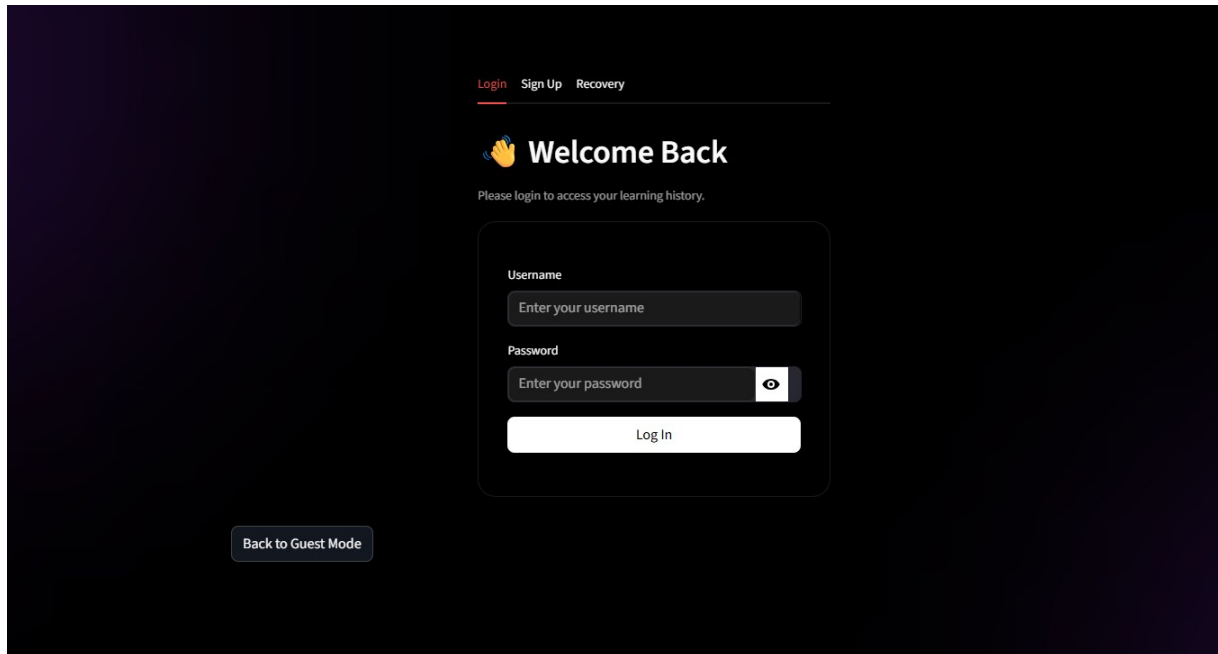
- Concept explanations generated by the Teaching Agent.
- Quiz or assessment results created by the Assessment Agent.
- Personalized learning recommendations suggested by the Recommendation Agent.
- Student progress tracking summaries from the Monitoring Agent.
- Final structured responses combining outputs from multiple agents.

Typical system screenshots may include:

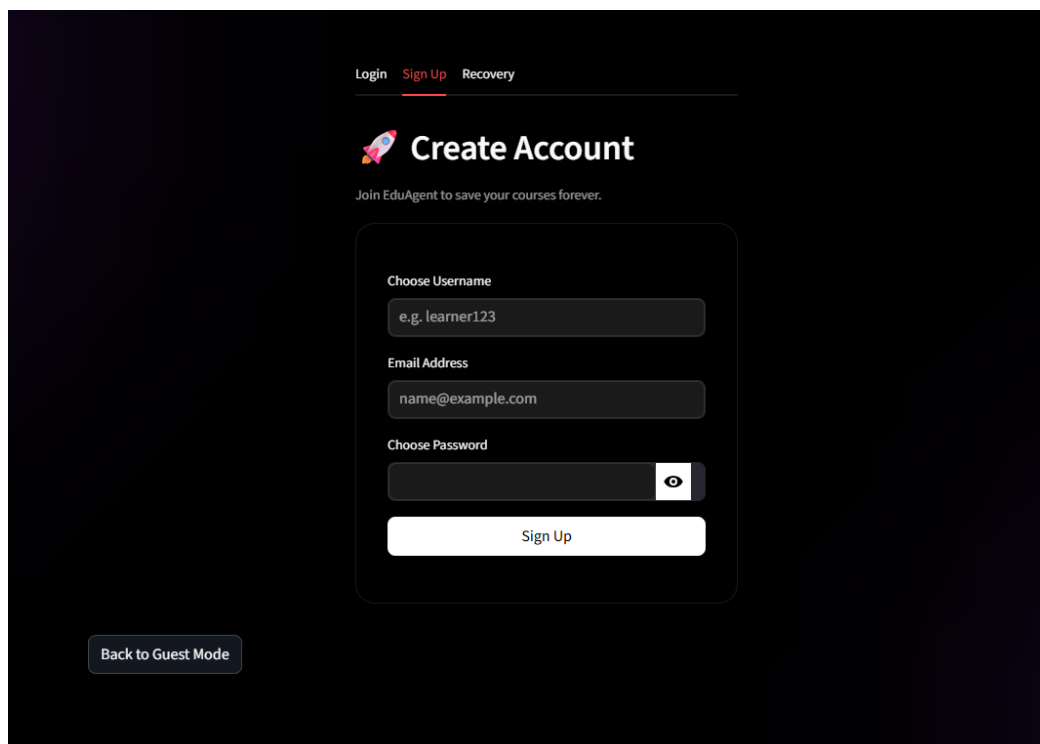
- Student query submission interface.
- Agent response showing concept explanations.
- Assessment or quiz evaluation results.
- Recommended learning path display.



- Generated progress or feedback reports.



The screenshot shows a login interface with a dark background. At the top, there are links for 'Login', 'Sign Up', and 'Recovery'. The 'Login' link is highlighted. Below the links is a 'Welcome Back' message with a hand icon. A message states 'Please login to access your learning history.' The login form contains fields for 'Username' (placeholder: 'Enter your username') and 'Password' (placeholder: 'Enter your password' with an eye icon for toggling visibility). A 'Log In' button is at the bottom of the form. A 'Back to Guest Mode' button is located at the bottom left of the form area.



The screenshot shows a 'Create Account' interface with a dark background. At the top, there are links for 'Login', 'Sign Up', and 'Recovery'. The 'Sign Up' link is highlighted. Below the links is a 'Create Account' message with a rocket icon. A message states 'Join EduAgent to save your courses forever.' The sign-up form contains fields for 'Choose Username' (placeholder: 'e.g. learner123'), 'Email Address' (placeholder: 'name@example.com'), and 'Choose Password' (placeholder: empty with an eye icon for toggling visibility). A 'Sign Up' button is at the bottom of the form. A 'Back to Guest Mode' button is located at the bottom left of the form area.

[Login](#) [Sign Up](#) [Recovery](#)



Reset Password

Don't worry, it happens to the best of us.

Username

New Password



Update Password

Back to Guest Mode

Deploy

 Inventions of Leonardo da Vinc



Course Material

 **Leonardo da Vinci: The Polymath Inventor**

 **Learning Objectives**

- Understand the concept of Renaissance humanism and its influence on Leonardo da Vinci's work
- Learn about the key concepts and technical explanations behind da Vinci's inventions, including flying machines, armored vehicles, and mechanical devices
- Analyze the real-world applications of da Vinci's inventions and designs in modern industries, such as aviation, military, robotics, and medicine
- Develop an appreciation for the interdisciplinary approach of da Vinci's work, combining art, science, engineering, mathematics, and anatomy
- Recognize the significance of da Vinci's anatomical studies and detailed drawings in informing his inventions and art

 **Module 1: Foundational Principles & Context**

Deploy

 **Concept Flashcards**

Renaissance Humanism

A cultural and intellectual movement that emphasized the potential of human beings to achieve great things through education, reason, and individualism, influencing Leonardo da Vinci's work and inventions. This movement played a significant role in shaping da Vinci's approach to art and science.

Flying Machines

Leonardo da Vinci's designs for flying devices, such as the flapping-wing ornithopter and the glider, which were based on the study of bird flight and the movement of wings. These designs demonstrate da Vinci's understanding of aerodynamic principles and his ability to apply scientific principles to real-world problems.

Mechanical Devices

The use of gears, pulleys, and levers in da Vinci's inventions, which demonstrate his understanding of mechanical advantage and his ability to analyze complex problems and develop innovative solutions. These devices were used to create automated systems and showcase da Vinci's ability to merge artistic and scientific principles.

Anatomical Studies

Da Vinci's detailed drawings of the human body, including the skeletal and muscular systems, which demonstrate his fascination with the inner workings of the human machine and informed his

Interdisciplinary Approach

The combination of art, science, engineering, mathematics, and anatomy in da Vinci's work, which has had a lasting impact on the development of modern technology and demonstrates the

 **Knowledge Check**

▼ Take Quiz

1. What movement influenced Leonardo da Vinci's work and emphasized the potential of human beings to achieve great things through education, reason, and individualism?

Select an option:

☒ A) Renaissance humanism

☐ B) Industrial Revolution

☐ C) Enlightenment

☐ D) Romanticism

Submit Answer 1

2. What was the inspiration behind Leonardo da Vinci's designs for flying machines?

Select an option:

☒ A) The study of bird flight and the movement of wings

☐ B) The observation of water and the behavior of fish

☐ C) The analysis of mechanical devices and gears

☐ D) The examination of human anatomy and the skeletal system

Submit Answer 2

3. In which modern industries can the practical applications of Leonardo da Vinci's inventions be seen?

Select an option:

☒ A) Aviation, military, and medicine

☐ B) Robotics, automation, and transportation

2. Key Outcomes:

- **Performance Optimization:** By parallelizing image generation, we reduced total course generation time from 75 seconds to 18 seconds.
- **Semantic Accuracy:** Using Vector Search ensures that if a user searches for "Red Planet," the system retrieves existing data for "Mars" with 90%+ similarity.
- **UI/UX Excellence:** The "EduVision AI" branding and Purple-Glow theme provide a state-of-the-art educational environment.

Conclusion:

The EduVision AI project marks a significant leap from simple chatbots to complex Agentic Systems. By distributing tasks among specialized agents and leveraging high-speed parallel processing, we have created a tool that not only provides information but creates a personalized, multimodal learning journey. The integration of Upstash Cloud DB ensures that this intelligence is scalable and persistent, paving the way for a new era of AI-driven education.

Future Scope & Enhancements:

Core Logic: Python 3.11 with LangChain for agentic chains.

LLM Engine: Groq Cloud (Llama 3.3 70B) for 10x faster reasoning compared to standard GPT models.

Vector Engine: Upstash Vector DB (Serverless Cloud) for semantic indexing.

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